Forest Rail

The Jingzhang Railway Heritage Park already left the rails in the city: people run, push strollers, walk along the tracks. This proposal does not reinvent that park, and does not turn it into a tech showcase. It asks something narrower — how the Dazhongsi end connects into the park, where people go after exiting the metro, where they can stop, whether they can sit down and work quietly, how seats and light strips can be safely AI-regulated, and how daily essentials can enter public space without a brand-store matrix [source:AGENT-TASKBOOK].

The answer is Forest Rail:

• Forest: Treat the Dazhongsi section as a park — manicured lawn, spaced trees, canopy above rooftops. Look out and see trees, not the glass wall opposite. The park is where people quiet down; human–nature relationship comes first.
• Rail (two tracks, separated): Jingzhang heritage rails are the urban clue. Decorative guiding rails give direction; AI operating rails run autonomous shuttles for passengers + automatic cargo configuration, safety first; physically separated from walking and decorative rails.
• Forest Rail terminal: intelligently regulates all public open seats / shared workstations, public ground light strips / low-illuminance edge lights, and daily-essentials supply at unbranded points (open/close, restock, rebalance). No face recognition, no brand ads.
• Unbranded daily-essentials trial points: do not start with each brand opening its own storefront in every direction. Instead, like old food shops — unified fronts, no brand logos; trial many sites; AI algorithms find where fit is highest and where one person's daily essentials can be met; high-fit nodes later open formal stalls; elsewhere use community quick-turnover shops, with AI inventory and autonomous cargo delivery.

> One line: rails are the city's clue; the park lets people quiet down; rusted rails point the way; autonomous shuttles put safety first; seats, light strips, and daily essentials can be smart-regulated — look out and see nature.

This proposal carries forward the Jingzhang Railway spirit of daring breakthrough — turning heritage rails into walkable, sittable, shared public infrastructure — without empty empowerment slogans or viral landmarks.

What this proposal refuses: district-wide building control, viral landmarks, light shows, planting strips posing as parks, private pods disguised as shared desks, decorative rails treated as climbing gear, "smart" traded for surveillance, brand-store matrices flooding the park.

Only one rail-guided corridor is added (guiding rails primary; operating rails only on isolated conceptual segments). Outside it, the existing street network stays.

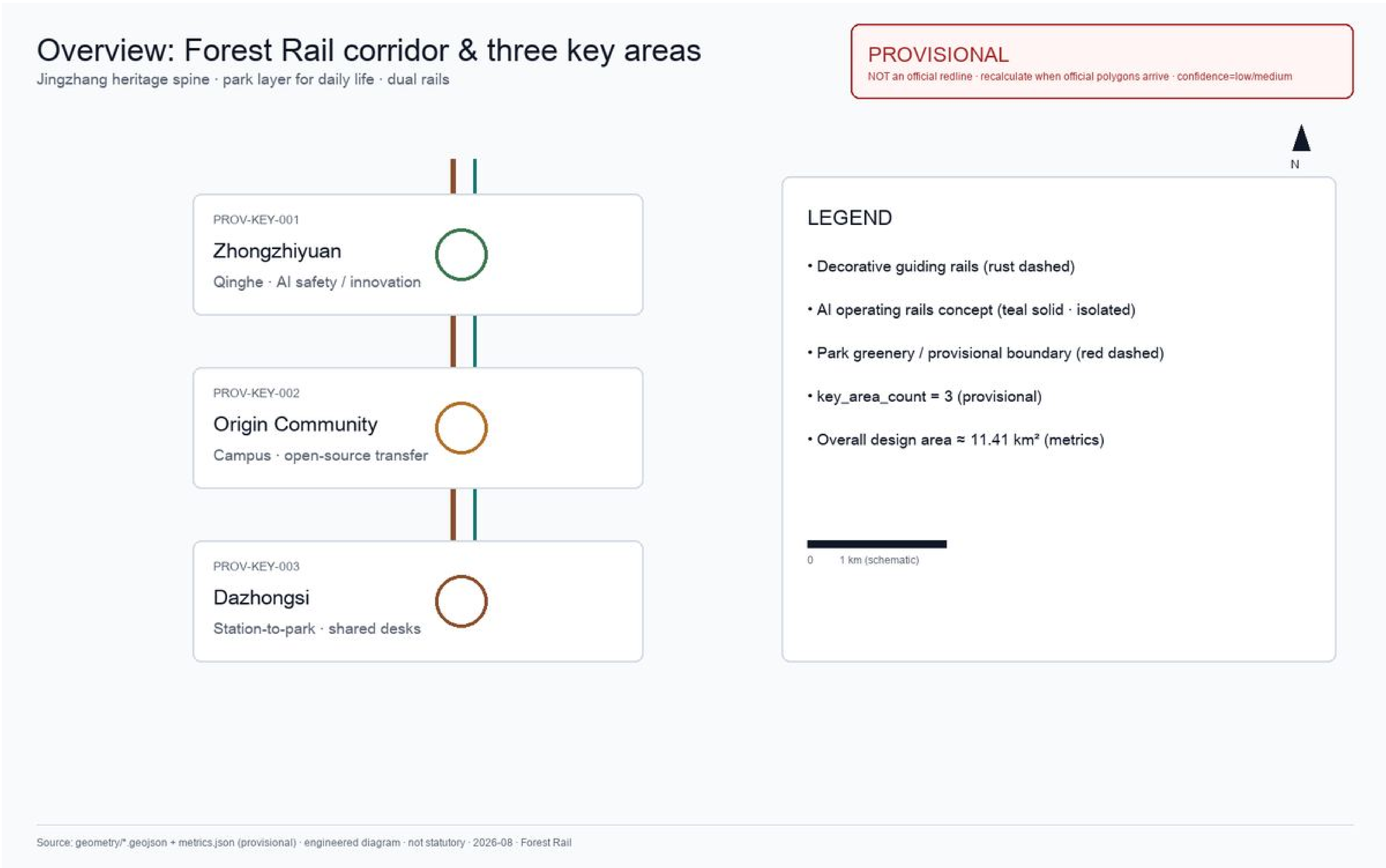
Design Basis and Source List

This proposal follows the Pre-qualification Announcement for the Centennial Jingzhang AI Innovation Belt Urban Design International Open Call [source:OFFICIAL-ANNOUNCEMENT], with machine-readable references from `brief/site-package/` [source:SITE-PACKAGE].

Current spatial data uses provisional boundaries (`geometry/site_boundary.geojson`, `geometry/key_areas.geojson`), labeled `provisional_constraint`, `official_boundary=false`. All metrics and drawings must be recalculated when official boundaries are released. This limitation does not block content scoring, but spatial conclusions in this text are for design discussion only.

The source registry (`data/source_registry.json`) lists 7 formal sources, 1 background source, and 1 provisional-only source. Background and provisional sources are not elevated to statutory authority [source:SOURCE-REGISTRY].

Authority order in this package: GeoJSON → metrics → matrices → sources/assumptions/self_check → proposal.md → figures → HTML → PDF. Concept renders do not replace evidence diagrams.



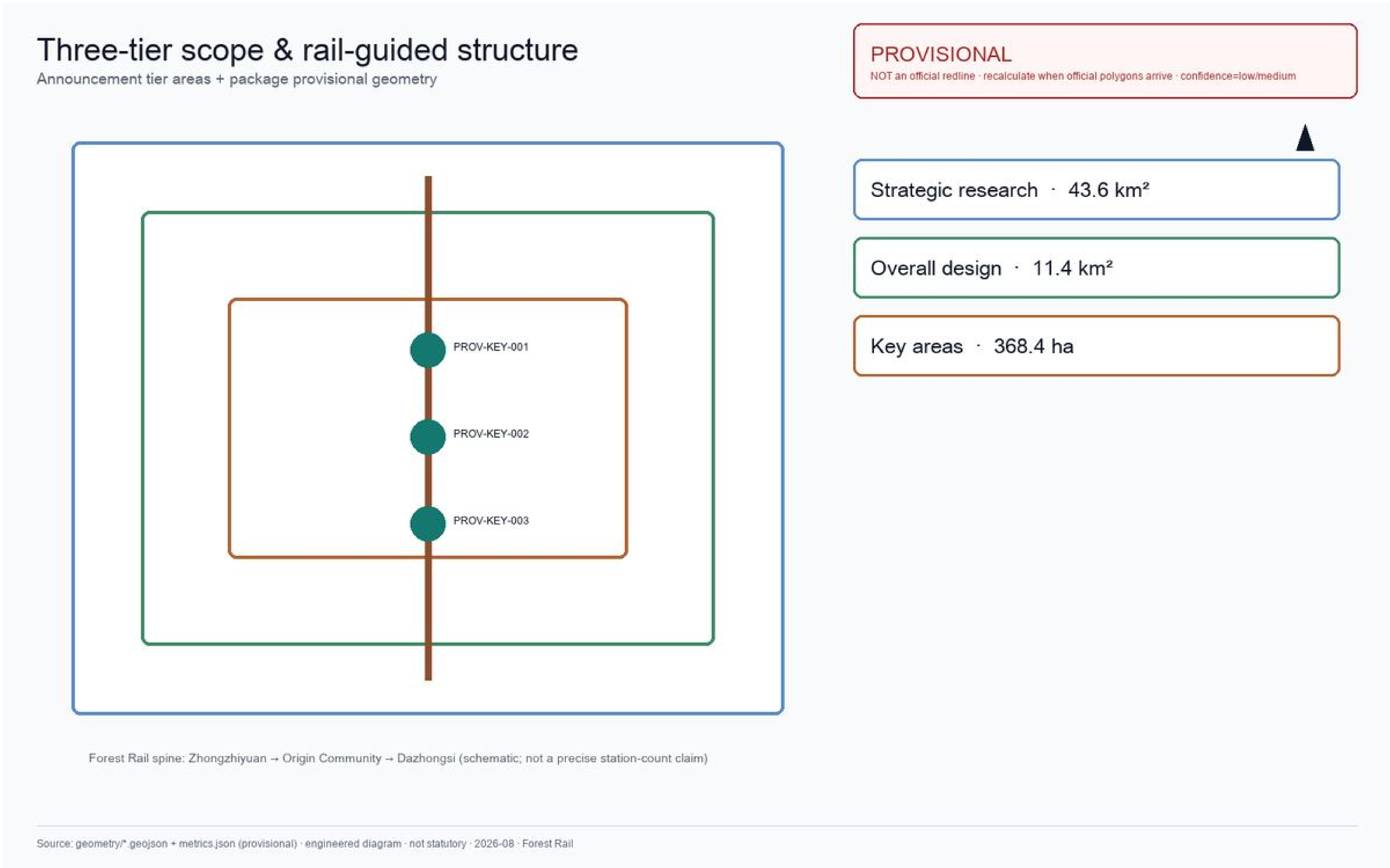
Source evidence chain and submission package diagram

Three-Level Scope Framework

The proposal follows the three tiers specified in the announcement [source:SITE-PACKAGE]:

Tier	Area	What Forest Rail does here
Strategic research (43.6 km²)	AI industry ecosystem and urban morphology	Confirms the Jingzhang corridor as spine; Forest Rail can extend north from Dazhongsi
Overall design (11.4 km²)	Urban renewal around the heritage park	Rail corridor links three key areas; park walking and soft paving; canopy as character control
Key area design (368.4 ha)	Three detailed design zones	Dazhongsi completes the park set: canopy, rail path, soft paving, shared desks, rust

Tiers are linked: strategy sets direction, overall design lands layers, key areas test whether components fit parcels.



Three-level scope and rail-guided corridor

Coordinated Research Area: Industry and Future City Research

The Jingzhang Railway is the narrative already embedded here. It opened in 1909; after going underground in 2019, the surface alignment became a heritage park corridor. The announcement requires integrating centennial Jingzhang culture, Zhongguancun innovation culture, and AI innovation culture [source:AGENT-TASKBOOK].

Forest Rail's judgment is narrow: do not invent a new story — use the rails; do not invent a new green corridor — connect the park to everyday Dazhongsi. Rails stay on the ground and rust; people walk soft paving between them. The line connects three key areas — Zhongzhiyuan by Qinghe, Origin Community near universities, Dazhongsi by the metro. The rail-guided corridor is the spine, not decoration.

Industry links follow routes people actually walk:

• University sourcing: work moves south along the heritage corridor, not floating circles
• Open-source collaboration: Origin Community hosts release and code work
• Enterprise translation: Dazhongsi hosts agents, terminals, and data-element firms
• Public services: ask-a-person nodes along the rail path, not unmanned screen walls
• International visits: walkable rail-path itineraries, not parade stages

Naming: Forest Rail is both title and spatial description. International line: *Forest Rail — the park layer of an AI city*. Rust + canopy green as visual motif; no full logo system — the track is the mark.

Urban Parks and Linear Parks: Global / Domestic Cases (Transfer Boundaries)

At the strategic tier, Forest Rail does not invent a new green-corridor story. It aligns transferable mechanisms from well-known urban and linear parks to the Jingzhang heritage corridor. Cases below are publicly built or operating projects. This proposal learns publicness and linear organization — not viral installations or climbable heritage consumption.

Case	Transferable mechanism	Boundary for this proposal (learn / do not learn)
New York Central Park	Multi-entry + layered walking; park as a public right	Learn: treat Dazhongsi as a real park (open lawn / spaced trees / seating); Do not: mega single-greenfield or closed heritage management
New York High Line	Abandoned rail → linear walk; keep rail artifacts as memory	Learn: rail-reuse narrative; Do not: elevated tourist consumption or climbable rails; our decorative rails ≠ climbing — soft paving for feet
Philadelphia Rail Park	Phased opening; 10–15 minute community reach	Learn: one-block-at-a-time lightweight pilots; Do not: sell unfinished segments as finished, or copy heavy elevated steel
London King's Cross public realm	Station squares stitching walking and mixed uses	Learn: exit into public space (Dazhongsi four quadrants); Do not: mall-led plaza logic
Madrid Río	Infrastructure gives way to continuous riverside green	Learn: stitch blocks split by infrastructure; Do not: tunnel-scale cost and schedule
Seoul Cheonggyecheon	Uncover water/green; everyday linear civic use	Learn: linear publicness; Do not: pumped faux-ecology or massive excavation narratives
Shanghai Xuhui Runway Park	Transport relic → park beside street	Learn: clear remnant-to-park narrative; Do not: wide hard runway feel or photo-booth stacking
Beijing Olympic Forest Park	Urban green heart; all-ages lawn and canopy	Learn: urban-forest character and inclusion; Do not: national-axis scale or a new mega-park land take
Jingzhang Railway Heritage Park (existing)	Rails already in the city; "three paths + green" walking	Learn: connect into Dazhongsi as incremental link; Do not: redo the park or invent a parallel corridor

Stance: carry forward Jingzhang daring — move heritage rails from "to be seen" toward "to guide, to share work, to host conceptual short-haul" — with the breakthrough landing on safety separation and everyday publicness, not empowerment slogans or viral gear.

Park case background sources are registered in `sources.json` (not statutory). North American urban/linear parks: [source:CASE-CENTRAL-PARK], [source:CASE-HIGH-LINE], [source:CASE-RAIL-PARK]. Europe and East Asia: [source:CASE-MADRID-RIO], [source:CASE-CHEONGGYECHEON]. Domestic and local baseline: [source:CASE-XUHUI-RUNWAY], [source:CASE-OLYMPIC-FOREST], [source:CASE-JZ-HERITAGE-PARK].

Urban life organism narrative · international motif (agent.5)

International claim (A): *Forest Rail — the park layer of an AI city.*

For the urban life organism: park/forest is a regulator, not decoration — spaced trees, lawn, and soft paving restore calm; calm enables creativity and green daily life. Spatial carriers are shared desks and rest areas inside a city park, not an office lobby moved outdoors.

Jingzhang three acts (daring breakthrough without tech-show spectacle):

1. 1909 — dare first: Jingzhang opens — engineering makes the impossible passable. 2. 2019 — rails stay in the city: after going underground, the surface becomes a heritage park — rails enter everyday walking. 3. This proposal — Forest Rail everyday: connect Dazhongsi into the park; rusted rails cut wayfinding friction; autonomous shuttles + Forest Rail terminal cut cargo / seat / light / daily-goods friction — design solves transport and supply friction, safety first, fail back to walking and humans.

Visual motif (not a full logo system): rust rail + canopy green + quiet / breathe / create.

Global AI innovation ecosystem cases & eight-factor map (agent.2)

Park cases cover publicness; these are the taskbook AI ecosystem cases (background only; no invented investment totals) [source:CASE-AI-STANFORD-RP]–[source:CASE-AI-TORONTO-VECTOR]:

Case	Transferable	Learn / do not
Stanford Research Park	land·space·industry·talent	Learn university–land–talent pipe; Do not low-density land sprawl
London Knowledge Quarter	space·industry·data·scene	Learn station-district knowledge reuse; Do not copy institute bulk
Singapore one-north / LaunchPad	land·space·compute·data·scene	Learn district-as-testbed; Do not single SOE ops for multi-actor corridor
Shenzhen Model Camp	space·compute·data·scene·talent	Learn shared compute/data platforms; Do not LLM-only towers
Zhangjiang AI hub pattern	space·industry·compute·data·scene	Learn vertical industry×AI alliances; Do not single-sector admin monopoly
Paris Station F	space·scene·compute·industry	Learn rail-hall reuse density; Do not private mega-landlord model
Toronto Vector Institute	talent·industry·capital·space	Learn non-profit talent–industry hub; Do not replace a belt with one institute

Eight factors (conceptual, not commitments): land (negotiable public-space pool); space (trackside shared desks/rest); industry (Zhongzhiyuan safety, Origin open-source, Dazhongsi terminals); capital (mechanism only, no amounts); talent (near-campus residency); compute (edge nodes, no vendor lock); data (anonymous counts, no face); scenes (verification miles along the corridor).

Urban agent protocol: walking-gap detection (auditable)

Step	Action	Who	Misjudgment fix
1 Collect	Anonymous heat + topology gap rules	Terminal + road graph	No identity stored
2 Candidate	Algorithm list of suspected gaps	Algorithm	Low confidence ≠ auto-build
3 Verify	Field walk / photo check vs traffic & rights	Ops inspector	Mark false_positive; feed rules
4 Approve	Project reviewer decides retrofit list	Reviewer	Appealable; minutes filed
5 Correct	Re-check built segment; revert if fail	Ops + design	Stop on safety/heritage veto

Agents do not replace planning approval and do not output personal profiles.

Industry test scenarios (agent.3 · ≥3 · uniform fields)

Field	TEST-01 daily restock	TEST-02 seats/lights	TEST-03 isolated AV shuttle
Goal	Restock latency + fair siting	Min open seats & staffed window	Safety first: E-stop / yield / fail-stop
Site	Dazhongsi trial points	Shared desks + curve benches	Isolated operable segment
Data	Anonymous footfall, stockouts	Noise/climate/rain/density	Vehicle sensors, boundary, E-stop logs
Owner	Ops + human stall review	Ops + staffed window	Safety assessor + ops (concept)
Stop	Basic supply stripped from low-demand nodes; privacy incident	Min seats missed; forced QR	Collision risk / E-stop fail / breach
Human review	Weekly siting audit	Daily min-supply check	Sign-off before/after each trial

Deaf/blind dedicated shuttle remains a concept extension of TEST-03 — not commercial.

Overall Design: Rail-Guided Corridor and Urban Renewal

Corridor Structure

The rail-guided corridor follows the Jingzhang heritage park alignment, with Dazhongsi as the southern gateway. It is not a newly drawn boundary — it reuses the railway alignment. The park is already walked; this proposal mainly connects Dazhongsi Station into that park.

```
Zhongzhiyuan (N) ──rail── AI Origin Community (M) ──rail── Dazhongsi (S) │ │ Qinghe interface Campus-district walking link
```

Tracks along the corridor serve two purposes and are physically separated [source:AGENT-TASKBOOK]:

Purpose	Material	Function	Boundary
Decorative guiding rails	Rusted / replica rails / ground painting + soft paving	Direction: eyes follow rust; feet on timber or crushed stone; sittable, slow	No running or climbing on rails; no light strips on decorative rails; signage minimal (Mode A)
AI operating rails (concept)	Autonomous shuttles on isolated operable segments	Low-speed passenger carriage + automatic cargo/materials configuration; safety first (E-stop, speed limit, yield to walking, fail-safe stop)	Separated from walking and decorative rails; not commercially mature; subject to professional study

Forest Rail Terminal: Environment Regulation (Seats · Light Strips · Daily Essentials · Access)

The Forest Rail terminal is an environment and daily-essentials regulation system — not building-wide control, not ad screens:

Target	What AI does	Safety & privacy boundary
Public open seats / shared desks	Open/close and power by time, noise, temperature/humidity, rain, anonymous crowd density	No identity logging; prefer close/lower power when crowded or weather is bad; retain a minimum number of open seats and a staffed service window — AI must not shut off all public supply
Public ground light strips / low-illuminance lights	Brightness and zone on/off by daylight, time of day, foot traffic; edges and seating only	No light shows, projections, or colored chasing lights; fault defaults to off or ultra-dim; avoid glare that harms low-vision users
Node access	Quiet workstations, rain-shelter seats, appointment safety nodes by schedule	Offline fallback to human/paper rules; human override always available
Autonomous shuttles (concept)	Dispatch, auto cargo config (including restock for daily points), path yielding	Safety first: any collision risk, boundary breach, or E-stop failure → stop and restore walking priority
Unbranded daily-essentials trial points	Open/close, restock, and rebalance daily necessities by anonymous footfall, time of day, stockouts, and turnover; after multi-site trials, output fit scores	No brand logos, no face recognition, no ads; formal stalls only after human review; failed sites can be withdrawn; algorithms must not chase high footfall only — low-demand communities keep minimum daily supply and a human restock channel

Most stretches are decorative guiding rails only. A few isolated stretches conceptually retain AI operating capacity. Pedestrians have priority at crossings; operating rails yield. If operating rails fail or are never deepened, guiding rails, seat regulation, light-strip regulation, and daily-essentials regulation still stand on their own.

Accessibility and Inclusion (mobility-limited users first)

Forest Rail treats people with limited mobility as a hard constraint, not a retrofit:

Group	Space & facilities	Forest Rail terminal / AV boundary
Wheelchair users	Rail path and station links avoid slopes where possible; grade changes use gentle ramps (≤1:20) and handrails; continuous non-slip soft paving, no step breaks; turning space at entries and seating	AI seat closure must not remove all reachable seats; offline human guidance to the nearest accessible route
Deaf and blind users	Tactile guidance with low-glare edge lights; tactile/braille cues at key nodes (not ad screens)	Concept reserved: dedicated AI rail shuttle — low speed, fixed stops, in-vehicle tactile guidance + voice/text dual-mode announcements + human help button; separated from freight/general passenger pods; safety first
Cognitive disability & low digital literacy	No app required to sit, shelter, or buy water; paper/icon rules and staffed help always available	AI suggests only; no forced QR codes; complex actions always have human fallback
Non-Chinese speakers	Safety and open/close information in Chinese and English; icons over long copy	Bilingual terminal and on-site signage; no face recognition, no ads

Fair siting: algorithms for unbranded daily points and seat open/close must include a minimum service radius for low-demand communities — high-footfall nodes may get more, but basic daily goods and seating cannot be stripped from low-demand nodes. All automated suggestions must be auditable, appealable, and human-overridable.

Land Use, Building Scale, and Retain-Renovate-Demolish Strategy

Renewal within the overall design scope falls into three categories:

- Retain: Existing quality buildings along the heritage corridor — unchanged
- Renovate: Buildings needing ground-floor program and facade adjustment — remove LED screens, increase under-canopy transparency
- Renew: New construction on identified low-efficiency sites — building height controlled below tree canopy

Land use classification follows national standards [standard:MNR-LAND-USE-CLASSIFICATION-GUIDE], covering the design boundary without overlap [data:geometry/land_use.geojson#LU-001]. Building footprints and retain/renovate/renew classification are recorded in [data:geometry/buildings.geojson#BLDG-001].

Where FAR, building height, or road setbacks lack official data, they are marked as unknown. No numbers are fabricated.

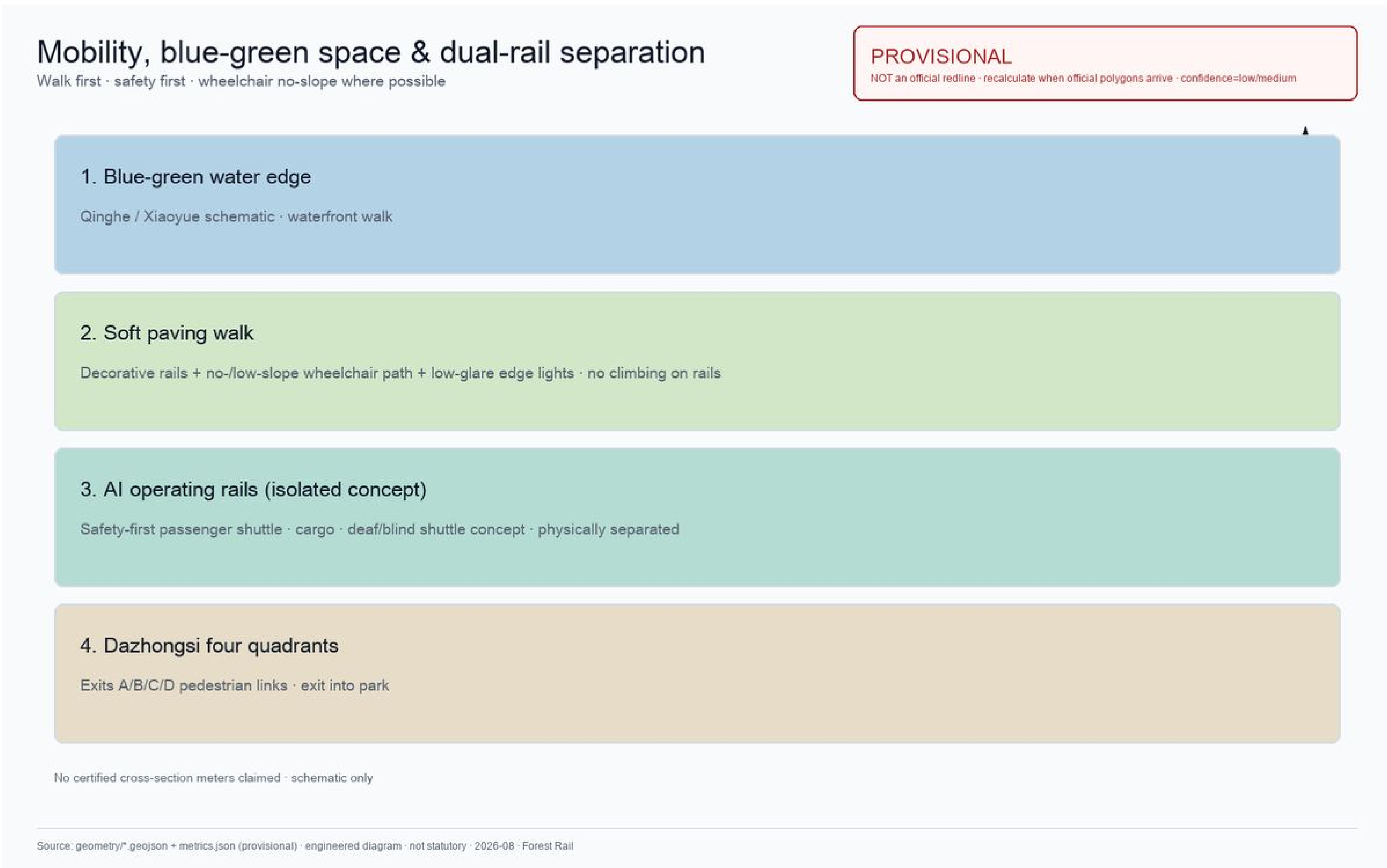
Transport, Rail, Municipal Infrastructure, and Public Services

Transport responds to the announcement's requirements for rail station integration, road micro-circulation, walking network gaps, and green transport [depth:traffic_rail_slow_parking].

Key interventions:

1. Dazhongsi Station four-quadrant pedestrian connection: Metro Line 13 Dazhongsi Station — establish walking connections across all four quadrants of the intersection, with rail paths extending from station exits into the district 2. North Fifth Ring crossing node: Pedestrian linkage where the heritage corridor crosses the North Fifth Ring Road (underpass or overhead facility) to maintain north-south continuity 3. Rail path / municipal road intersections: Pedestrians have priority at crossings; rail tracks embedded in road surface indicate direction; no traffic signals govern the rail path

Road and walking network layers remain within the submission boundary [data:geometry/roads.geojson#ROAD-001], cross-checked with public space and green space. With a provisional boundary, transport conclusions serve as design discussion only.



Transport, walking, and blue-green public space composite system

Municipal infrastructure covers edge-computing nodes, distributed energy, and traditional utility integration. Missing utility, energy, and drainage data are listed as prerequisites for detailed design, not fabricated.

Blue-Green Network, Public Space, and Urban Character

Blue-green space follows the Jingzhang heritage corridor as its spine, coordinating Qinghe River, Xiaoyue River, and surrounding community movement [depth:blue_green_public_space].

The tree canopy layer is the primary character control mechanism: native trees with crown spread ≥8m (Chinese Scholar Tree, ash, ginkgo, goldenrain tree), targeting ≥70% coverage of building rooftops. Trees are not pruned into geometric shapes. Building height below the canopy — this is the basic criterion for the urban forest.

Character principles:

- Building facades use materials that age: brick, concrete, wood. No smooth glass curtain walls
- Rust and grey-green are the base colors, derived from oxidized rail and tree canopy
- Rooftops are for greening or equipment, not sculptural form
- Public art accepts only site-history-related, aging-compatible works — iron, stone, wood. No stainless steel or lighting installations

Which elements are design recommendations versus those pending official confirmation is itemized in `standard_matrix.json`.

Key Area Detailed Design

Zhongzhiyuan AI Autonomous Innovation Accelerator

Location: Northern end, along Qinghe River [data:geometry/key_areas.geojson#PROV-KEY-001].

Position: Garden-type full-stack autonomous innovation district. The Qinghe interface is its public living room — combining green space, stormwater management, walking and cycling, and AI safety testing display.

Spatial actions:

- Qinghe interface: Set back buildings, create under-canopy walkways and waterfront grass slopes
- Industry display: Standards development, safety evaluation, model red-team testing translated into visitable collaboration nodes — by appointment, not as open attractions
- External transport: Strengthen walking connections to areas north of the Fifth Ring
- Low-carbon energy: Distributed PV combined with edge computing, as an infrastructure prototype for further development

Beijing AI Origin Community

Location: Middle section, near Tsinghua, Beihang and other universities [data:geometry/key_areas.geojson#PROV-KEY-002].

Position: University-adjacent technology transfer and talent community.

Spatial actions:

- Campus-park-district walking integration: Break through walls and gaps, connect campus gates to park entrances via rail paths
- Launch space: Open-source release hall, public code wall (physical screen displaying real-time open-source contributions), small presentations
- Talent services: Housing, dining, sports embedded in the streetscape, not concentrated in a "talent apartment complex"
- Open-source collaboration: 24-hour open collaboration spaces, divided into quiet zones and discussion zones by usage intensity

Dazhongsi AI Industry Cluster — Urban Forest

Location: Southern end, adjacent to Metro Line 13 Dazhongsi Station [data:geometry/key_areas.geojson#PROV-KEY-003].

Position: Urban intelligent economy district, and the fullest end of Forest Rail — exit the station into a park, and find a shared desk in that park.

Five components:

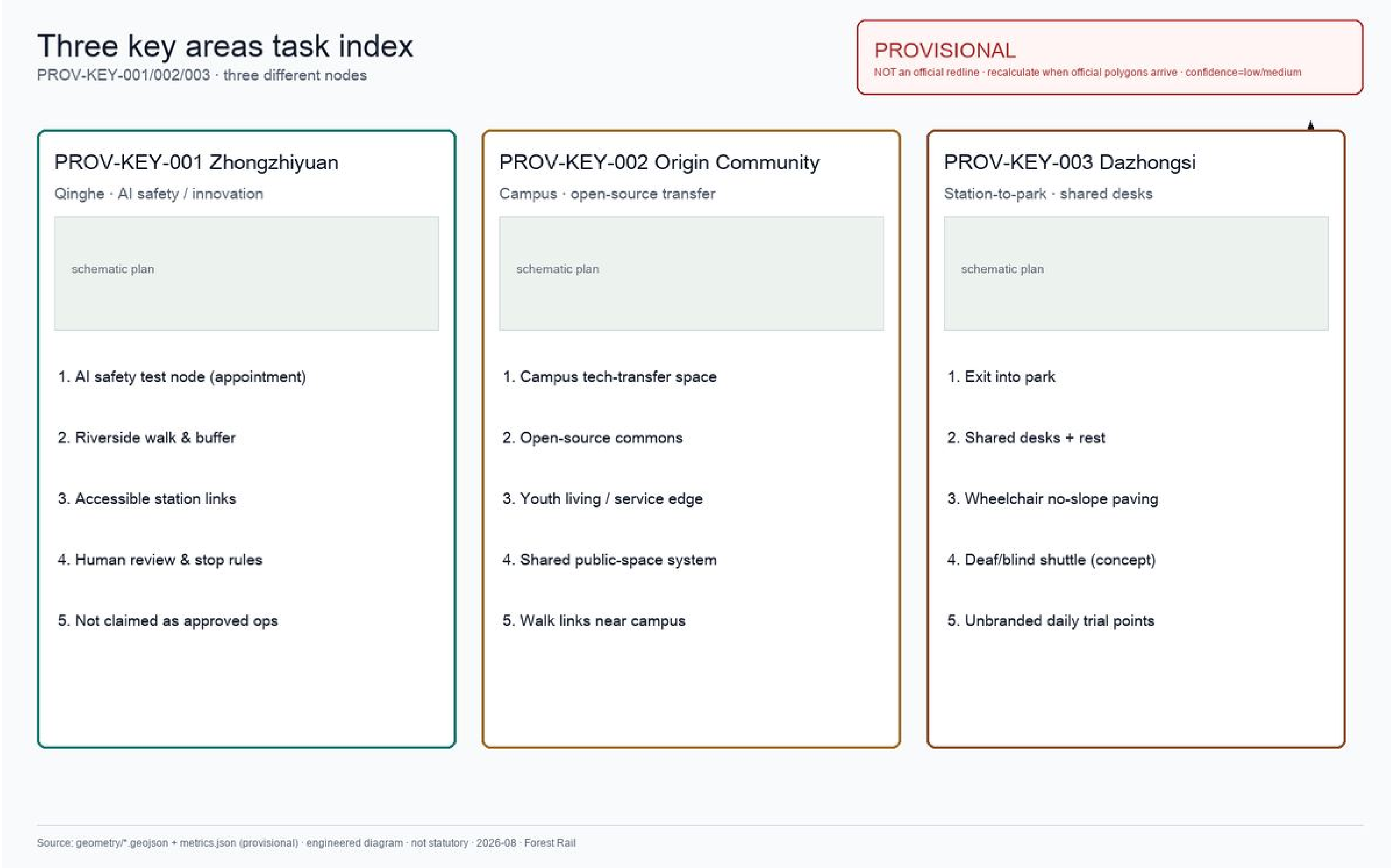
Component	How it is used in Dazhongsi	What is not allowed
Canopy layer	Chinese Scholar Tree and ash as primary species, crown spread ≥8m, covering existing building roofs and new low-rise structures	No geometric pruning; no ornamental small trees as substitutes
Rail path	From Dazhongsi Station exits, decorative guiding rails (rusted / replica / ground painting) point into the park; people walk soft paving	No elevated landscape bridges; no polishing; no painting; no light strips; no running or climbing on rails
Soft paving	Between-rail walkways use timber boardwalk, permeable brick, or crushed stone; comfortable underfoot; wheelchair routes avoid slopes and stay flush with walking paths	No large-area asphalt; no polished stone (slippery in rain); no step breaks
Under-canopy workspace	Dedicated shared workstation precinct in the park: many desks, glass partitions, hot-desks anyone can use; building ground floors also shaded by canopy	No private assigned pods; no LED facades; no viral photo-booth cabins
Rust	Rail tracks and steel elements maintain natural oxidation as time markers	No Cor-Ten imitation rust panels; no artificial aging with clear coat

Dazhongsi Station integration: From exits A/B/C/D, a rail path enters each quadrant. Emerging underground, the first view should be park and rusted rail — not ad screens.

Commercial services (unbranded daily essentials → formal stalls by fit): Do not start with each brand opening its own storefront in every direction. Instead, set unified, unbranded daily-essentials trial points (water, simple meals, basic daily goods — what one person needs in a day), like old food shops — unified fronts, no logos, no marquees. After multi-site trials, AI algorithms (anonymous footfall, time-of-day demand, turnover, stockouts) judge where fit is highest and where daily essentials can be met; high-fit nodes later open formal commercial stalls (open admission, not AI anointing a brand); elsewhere use community quick-turnover shops, with Forest Rail terminal AI managing inventory and autonomous shuttles delivering restocks. Low-demand communities keep minimum daily supply — algorithms must not withdraw a point only because footfall is low. Company showrooms may face the rail path but must not overpower the park's quiet character; data-element services require compliance, authorization, and auditability.

Park integration: The rail path sits in a park — manicured lawn, spaced trees, soft paving. Shared desks and daily points both serve the park: rails are the urban clue; the park is where people quiet down and human–nature relationship comes first — not an office lobby or brand street moved outdoors.

Concept views (not formal evidence figures; spatial claims remain in GeoJSON / metrics / the five evidence diagrams):



Three key areas index and design tasks

All three key areas appear in `geometry/key_areas.geojson`. Current data is provisional constraint; the proposal, HTML, and self_check state that it cannot serve as approval or scoring basis.

A Day Here: Personas and Scenarios

Five User Personas

Persona	Who	What they do on Forest Rail	Data boundary
Walking commuter programmer	AI engineer living nearby, walks to work daily	Exits Dazhongsi Station, walks 800m along the rail path to the office; eats lunch under the canopy; walks back along the rail path after work	No personal commute tracking; rail path foot traffic is anonymous counting only
Parent walking with child	Resident of surrounding neighborhoods	Pushes a stroller along the soft paving in the evening; child runs on soft paving beside the rails (not on the rusted rails); sits on under-canopy benches chatting	No family profiling; no commercial push notifications
Corporate visitor	Business guest of a major company	Arrives at Dazhongsi Station, rail path guides to company showroom; walks along rail path to next node after the meeting	Company logos and cases require rights clearance
Solo freelancer	Person without a fixed office	Finds an open shared desk in the park workstation precinct; opens when quiet, closes when crowded; hot-desk, no reserved seats	No recording of workstation user identity
Retired daily walker	70-80 years old, lives nearby	Walks along the rail path every morning, sits on the bench at the curve watching trees; pavement must be non-slip	No health data collection
Wheelchair user	Person who uses a wheelchair	Uses continuous soft paving on no-slope / low-slope segments; accessible seats and turning space at station–park links	No mobility profiling
Deaf or blind user	Visitor or resident with hearing or vision loss	Moves by tactile guidance, low-glare edge lights, and (concept) dedicated AI rail shuttle	No face recognition; shuttle offers voice/text/tactile modes and on-demand human help
Non-Chinese visitor & low digital literacy	International guest or resident who avoids apps	Reads icons and bilingual signs to sit, shelter, and buy water; staffed help for complex steps	No forced registration or QR codes

Scenario Cards (≥10)

#	Scenario	Spatial carrier	What happens	Who uses it
01	Exit station, enter park	Dazhongsi Station exits	Decorative guiding rails point to park entries; canopy provides auxiliary shade; rain relies on shelters/arcades as backup — not light shows	All arrivals
02	Rail path walking	Entire Dazhongsi rail path	Eyes follow rusted rails; feet on timber or crushed stone. Speed naturally slows. No destination signs urging you forward	Commuters, walkers
03	Shared under-canopy workstations	Dedicated park precinct with glass-partitioned multi-desk bays	Shared hot-desks: sit if free. Forest Rail terminal opens/closes seats and power by noise / climate / rain / crowd	Freelancers, lunch-breakers, passersby working briefly
04	Rusted rail curve bench	Rail path turning points	Benches on the outside of curves; sit and watch the rusted rail and trees. No signage; discover by sitting; terminal may schedule that seat group	Elderly, parents with children
05	Company showroom along the rail	Dazhongsi ground-floor retail	AI agent and terminal company display spaces open toward the rail path; visible as you walk past	Visitors, passersby
06	Rainy day under-canopy shelter	Dense canopy segments + node shelters	Canopy as auxiliary shade; heavy rain backed up by artificial shelters / semi-open arcades; AI prompts open seating and shelter nodes	Everyone
07	Open-source launch hall	Origin Community	Code contribution display and small presentations for universities and open-source communities. Physical screen scrolling real-time commits	Developers, university faculty and students
08	Inter-node autonomous shuttle (concept)	Isolated operable segments	Safety-first autonomous vehicles for passengers + automatic cargo; E-stop / speed limit / yield; separated from walking and decorative rails	Concept verification stage, not mixed daily use
08b	Deaf/blind accessible shuttle (concept)	Fixed stops on isolated operable segments	Dedicated AI rail shuttle: tactile guidance + voice/text announcements + human help button; low speed, fixed stops; separated from freight pods	Deaf/blind users, people needing assisted movement
15	Wheelchair-accessible walking	Dazhongsi four-quadrant links + no-slope rail segments	Flush station links; continuous non-slip paving; gentle ramps + handrails where unavoidable; AI seat closure retains reachable seats and human guidance	Wheelchair users, parents with strollers
16	Minimum public supply	Low-demand community nodes along the corridor	Minimum open/close floor for unbranded daily points and public seats; high footfall must not remove the last water/seat/staffed window	Low-demand residents, low digital-literacy users
09	Night light-strip regulation	Soft paving edges along the rail path	Terminal regulates public ground light strips / low-illuminance lights: edges and seating only. No light shows, no projections	Night runners, late returners
10	Qinghe innovation corridor	Zhongzhiyuan along Qinghe	Waterfront walkway connects to the rail path, stormwater grass slopes double as resting lawns. AI safety testing nodes open by appointment	Visitors, walkers
11	Data element salon	Dazhongsi district	Under compliance, authorization, and auditability, displays data element and digital asset circulation as an urban service interface	Enterprise clients, regulators
12	University tech transfer street	Origin Community	Incubation, display, legal, IP, and investment services organized along the rail path for university technology transfer	Startup teams, investors
13	Unbranded daily-essentials trial points	Multi-site fronts along the rail path	Unified food-shop-style fronts, no brand logos; AI regulates open/close, restock, and rebalance of daily necessities; multi-site trials output fit scores	Commuters, residents, passersby working briefly
14	High-fit nodes → formal stalls	Nodes with high fit after AI trials	Open formal commercial stalls (admission-based); elsewhere switch to community quick-turnover shops with AI inventory + autonomous cargo delivery	Community vendors, residents

All AI governance follows data minimization, open sourcing, explainability, and human review. Seat/daily-point open/close must retain staffed help and minimum public supply; siting algorithms must not chase high footfall at the expense of basic service in low-demand communities. Urban intelligent agents assist in identifying walking network gaps, public space activity, and facility maintenance needs but do not replace planning approvals or output unauthorized personal profiles.

Three Quiet Landmarks (Not Attractions)

No "check-in points" are designated here. The following are places you notice while walking — no signage, no wayfinding, no photo-taking recommendations:

1. Rusted rail curve with an old Scholar Tree: At a bend in the rail path, an existing large Scholar Tree. The curve makes you turn your head and see the tree. Unnamed, unmarked. 2. Rail path–Xiaoyue River junction, crushed stone square: Where the rail path crosses Xiaoyue River, a crushed stone surface forms a small square. Water, rusted rail, and canopy meet here. No sculpture. 3. Old platform retaining wall: If Jingzhang Railway-era platform remnants exist in the district (retaining walls, steps), they are kept as found. The wall surface is not cleaned; moss grows. A bench is placed beside it.

Renewal Project List and Phasing

ID	Project	Type	Key dependencies	Phase
FR-01	Dazhongsi Station four-quadrant rail path connection	Walking / rail integration	Rail station, intersection traffic organization	Near-term pilot
FR-02	Rail path wayfinding segment installation (Dazhongsi)	Public space	Rusted rail sourcing, soft paving materials, drainage	Near-term pilot
FR-03	Canopy layer planting (Dazhongsi)	Greening	Tree specification, soil improvement, utility avoidance	Near-term start, 3-5 year canopy maturity
FR-04	Under-canopy workspace ground floor renovation	Building renovation	Ownership, ground-floor program adjustment	Medium-term
FR-05	Zhongzhiyuan Qinghe innovation interface	Blue-green / industry	River blue line, ecology and flood control conditions	Medium-term
FR-06	Origin Community campus walking integration	Walking	Campus boundary, wall removal negotiation	Medium-term
FR-07	Inter-node AI operating rail concept segment (passenger short-haul + automatic cargo)	New infrastructure (concept)	Feasibility, safety assessment, separation; see PR-12	Long-term / pending
FR-08	Full rail path connection (three key areas linked)	Public space / walking	North Fifth Ring crossing, corridor ownership; see PR-03/05	Long-term
FR-09	Forest Rail terminal pilot (seats·lights·daily·access)	Ops / governance	Min supply, fair siting, staffed window; see PR-11/13/14	Near-term pilot

Near-term pilots can start with lightweight interventions (spread crushed stone, place benches, plant saplings) without waiting for all engineering conditions. Medium and long-term projects require confirmed regulatory plans, municipal engineering, and property rights.

Prerequisites checklist (actors, rights, cost band, acceptance, ops): `data/processed/prerequisites_checklist.csv` [source:PREREQUISITES].

Phasing spatial evidence [data:geometry/phasing.geojson#PHASE-001].

Metrics Framework

Metrics are divided into three categories:

Type	Examples	Data source	Precision
Spatial metrics calculable from submitted geometry	Boundary area, green ratio, public space ratio, building footprint	GeoJSON layers	Limited by provisional boundary
Control metrics requiring official regulatory plans	FAR, building height, building density, setbacks	Pending official plans	Currently marked unknown
Performance metrics requiring ongoing operational data	Rail path daily foot traffic, under-canopy workstation utilization, node access frequency	Post-operation collection	Currently design reference values

Three categories enter `metrics.json`, `assumptions.json`, and `compliance_matrix.json` respectively. Full calculations are in `metrics.json`; this text explains design intent only.

Metrics recalculation & evidence (provisional)

Numbers locked from metrics.json · zh/en must match

PROVISIONAL

NOT an official redline · recalculate when official polygons arrive · confidence=low/medium

Metric	Value	Confidence	Status / formula
site_area_sqm	11,412,825.386 m² (~11.41 km²)	low	provisional
green_ratio	0.123423	medium	provisional
green_area_derived	1,408,605.148 m²	medium	site×green_ratio
public_space_ratio	0.073281	medium	provisional
public_space_derived	836,343.257 m²	medium	site×public_space_ratio
building_footprint_area_sqm	310,807.184 m²	medium	provisional
key_area_count	3	low	provisional
floor_area_ratio	unknown	unknown	await official
avg_height_m	unknown	unknown	await official

Evidence chain: GeoJSON → metrics.json → matrices → proposal / HTML / PDF

Source: geometry/*.geojson + metrics.json (provisional) · engineered diagram · not statutory · 2026-08 · Forest Rail

Core metrics recalculation and evidence chain

Key spatial metrics can be verified against [metric:site_area_sqm] and [data:geometry/green_space.geojson#GREEN-001].

Compliance Matrix Coverage

`compliance_matrix.json` maps each requirement from Announcement 1.3-1.5 and agent.1-agent.6:

Requirement	Proposal section	Core evidence
1.3 World-class AI innovation ecosystem	Strategic research + Key areas	Innovation chain spatial layout, scenario cards
1.4 Three-tier scope	Three-tier scope	Task table per tier
1.5.3.1 Zhongzhiyuan	Key areas: Zhongzhiyuan	key_areas PROV-KEY-001
1.5.3.2 Origin Community	Key areas: Origin Community	key_areas PROV-KEY-002
1.5.3.3 Dazhongsi	Key areas: Dazhongsi	key_areas PROV-KEY-003
agent.1 Overall concept	Overall design: Corridor structure	Rail-guided corridor + three key area linkage
agent.2 AI full-stack innovation	Global AI ecosystem cases + eight factors	7-case table + factor map; CASE-AI-*
agent.3 AI+ scenarios	Personas + test cards	16 scenario cards + TEST-01/02/03
agent.4 Public space and landmarks	Dazhongsi urban forest + quiet landmarks	Five components + three quiet landmarks (not attractions)
agent.5 Cultural narrative integration	Urban life organism + Jingzhang three acts	International claim A; rust/canopy motif
agent.6 Global activities and long-term operations	Phasing + prerequisites	FR list; prerequisites_checklist.csv

Risk, Copyright, and Compliance

Provisional Boundary

All spatial conclusions are limited by provisional boundary status. When official boundaries are released: re-run scaffold, self-check, drawings, and HTML. Single-file replacement is not sufficient.

AI Operating Rail & Forest Rail Terminal (Concept)

• Autonomous shuttles: safety first — speed limit, E-stop, yield to walking and decorative-rail zones; stop on collision risk, boundary breach, or E-stop failure. May carry passengers and auto-configure cargo/materials pods (including restock for daily points). Conceptual suggestion, not a mature commercial claim.
• Forest Rail terminal: intelligently regulates public seats/shared desks, ground light strips / low-illuminance lights, and daily-essentials open/close/restock/rebalance at unbranded points; no face recognition, no brand ads; human override when offline; retains minimum open seats, staffed windows, and basic supply in low-demand communities.
• Unbranded daily-essentials trial points: start with no brand logos; AI outputs fit scores and supply suggestions only; upgrading to formal stalls requires human review and open admission; failed sites can be withdrawn; must not withdraw low-demand nodes for footfall alone.
• Accessibility: wheelchair routes avoid slopes where possible (unavoidable ramps ≤1:20 + handrails); dedicated AI rail shuttle for deaf/blind users reserved as concept (tactile + voice/text + human button); cognitive/low-literacy/non-Chinese users are not forced into apps — bilingual signage and staffed fallback.
• Decorative guiding rails: no climbing or running; physically separated from operating rails. Rails are the urban clue; the park prioritizes quiet and human–nature relationship.

Copyright & bilingual equivalence

• Rights chain: `report/copyright_statement.md` [source:COPYRIGHT-STATEMENT] (per-asset; evidence boards are engineered diagrams)
• Bilingual check: `report/bilingual_equivalence_checklist.md` [source:BILINGUAL-CHECKLIST]
• Figure footers cite package geometry/metrics + 2026-08 only — no hallucinated agencies, XX-city placeholders, or Working Draft seals

Missing Data

Gaps listed in `missing_data_checklist.csv` — official boundaries, regulatory plans, roads, municipal engineering, heritage protection — enter `assumptions.json` and self-check. Conclusions lacking official conditions are downgraded to pending confirmation.

Copyright (display)

The proposal provides bilingual versions (this file and `proposal.md`). Images, data, and code assets are documented in `sources.json` and `report/copyright_statement.md`. HTML pages do not load remote scripts, map tiles, fonts, iframes, or external APIs.

Disclaimer

This proposal does not claim official approval, regulatory plan status, final property rights, final construction scale, or implementation guarantee.

References

See [source:OFFICIAL-ANNOUNCEMENT], [source:AGENT-TASKBOOK], [source:SITE-PACKAGE], `sources.json`, `metrics.json`, `compliance_matrix.json`.
• brief/public-brief.md
• brief/site-package/design_brief.json
• brief/site-package/allowed_design_space.json
• data/processed/agent_fact_pack.md
• data/processed/project_scope_summary.csv
• data/processed/agent_task_requirements.csv
• data/processed/missing_data_checklist.csv
• Full source index: `sources.json`, `metrics.json`, `compliance_matrix.json`, `standard_matrix.json`, `design_depth_matrix.json`